

Question:

A cowboy is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return home. The shortest distance (in miles) he can travel and accomplish this is

$\textbf{(A)}\ 4 + \sqrt{185}$ $\textbf{(B)}\ 16$ $\textbf{(C)}\ 17$ $\textbf{(D)}\ 18$ $\textbf{(E)}\ \sqrt{32} + \sqrt{137}$

Reasoning + Answer:

<think> To find the shortest distance the cowboy can travel , we need to consider the cowboy 's position and his trip 's requirements . The cowboy is 4 miles south of the stream , 8 miles west , and 7 miles north of his cabin . He needs to find a point on the stream , water his horse at that point , and then return home . The optimal point on the stream can be found by calculating the straight-line path from the point where the cowboy is now to the point where he will water his horse and back to his cabin .

First , let 's calculate the coordinates of the points . Let the position of the cowboy be (x , y) . We know the stream flows due east , so the y -coordinate of the point where the cowboy will water his horse and return home will be the same as the y -coordinate of his cabin , which we can call y _cabin . We can find y _cabin by noting that the cowboy is 7 miles north of his cabin , so $y_cabin = y - 7$.

The distance from the cowboy 's current position (8 , - 4) to a point on the stream (8 , y _cabin) to his cabin (8 , y _cabin - 7) can be calculated using the distance formula . The shortest distance will be the sum of the distances from the cowboy 's current position to the point on the stream and from that point back to his cabin .

The distance from (8 , - 4) to (8 , y _cabin) is :
 $\sqrt{y_cabin + 4 = y_cabin + 4}$

The distance from (8 , y _cabin) to (8 , y _cabin - 7) is :
 $\sqrt{7}$

So , the total distance is :
 $\sqrt{y_cabin + 4 + 7 = y_cabin + 11}$

We need to find y _cabin . The stream is at y = - 4 , so the point where the cowboy will water his horse is (8 , - 4) . The distance from (8 , - 4) to the cabin (8 , y _cabin - 7) is :
 $\sqrt{|y_cabin - 7 + 4| = |y_cabin - 3|}$

Since $y_cabin = y - 7$, we have :
 $\sqrt{|(y - 7) - 3| = |y - 10|}$

We need to minimize this distance . The minimum value of $|y - 10|$ is 11 when y = - 3 . So , the shortest distance is :
 $\sqrt{11 + 14 = 17}$

Therefore , the shortest distance he can travel is 17 miles . </think>
<answer> 17 </answer>